SETUP	# FP	IMAGENET	IN-REAL	IN-V2	IN-A	IN-R	IN-SKETCH	CONFLICT STIMULI	IN-C	MEAN
Baselines										
Nominal training	×1	82.64%	87.33%	71.42%	28.03%	47.94%	34.43%	30.47%	64.45%	55.84%
Adversarial training	×1	76.88%	83.91%	64.81%	12.35%	55.76%	40.11%	59.45%	55.44%	56.09%
Fine-tuned MAE-B16	×1	83.10%	88.02%	72.80%	37.92%	49.30%	35.69%	27.81%	63.23%	57.23%
AdvProp	×1	83.39%	88.06%	73.17%	34.81%	53.04%	39.25%	38.98%	70.39%	60.14%
Pyramid-AT	×1	83.14%	87.82%	72.53%	32.72%	51.78%	38.60%	37.27%	67.01%	58.86%
Indep. networks ensemble	×2	82.86%	87.78%	71.73%	25.99%	54.20%	37.33%	46.41%	65.61%	58.99%
Individual networks ensemble	×4	81.31%	86.97%	70.21%	23.13%	54.82%	39.51%	56.02%	68.17%	60.02%
Fixed grid search on 1000 images										
Single soup	×1	82.49%	87.85%	71.99%	34.31%	53.84%	39.84%	38.52%	66.82%	59.46%
Dataset-specific soups	×1	82.29%	87.89%	71.95%	38.27%	56.39%	40.73%	67.03%	69.34%	(64.24%)

Table 1. **Comparison on IMAGENET variants:** we report the classification accuracy of various models on the IMAGENET variants, together with the number of forward passes they require. The soups are selected via a fixed grid search on the interpolation weights with 1000 points for each dataset. The last row shows the results of the best found soup for each dataset.

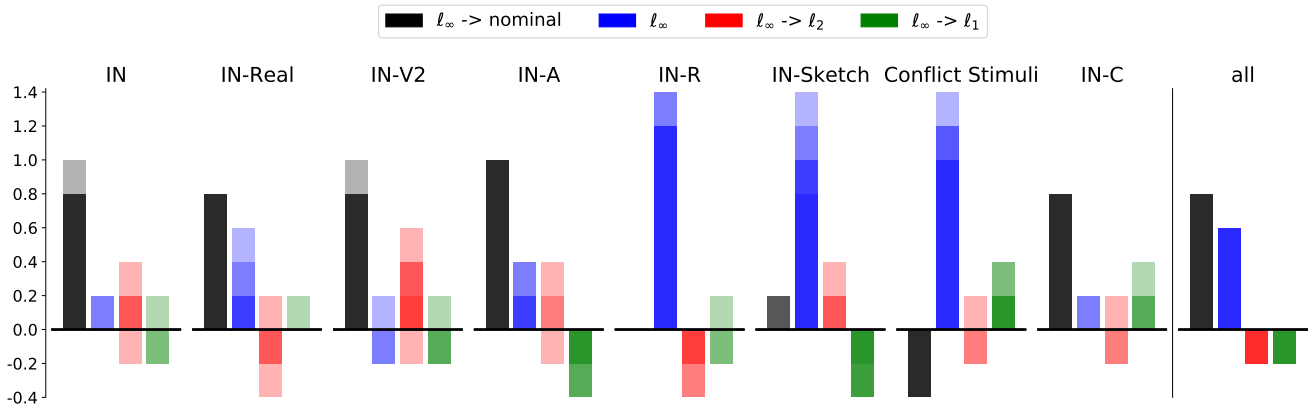

Figure 7. **Soup compositions on IMAGENET variants:** for each dataset we plot the composition of the 5 best soups, i.e. the linear weights for the individual models, as measured by grid search over 1000 points on weights in the range $[-0.4, 1.4]$ with granularity 0.2. Additionally we show the composition of the model achieving the best average accuracy over all 8 variants.

all, The soup selected for best average accuracy (across all datasets) outperforms all baselines, except for the ensemble of four models (with $4\times$ the inference cost), and AdvProp, which requires co-training of clean and adversarial points. These results show that soups with robust classifiers are a promising avenue for quickly adapting to distribution shifts.

Composition of the soups. To analyze which types of classifiers are most relevant for performance on every distribution shift, we plot in Fig. 7 the breakdown of the weights of the five best soups (more intense colors indicate that the corresponding weight or a larger one is used more often in the top-5 soups). First, one can see that the nominally fine-tuned model (in black) is dominant, with weights of 0.8 or 1, on IMAGENET, IMAGENET-REAL, IMAGENET-V2, IMAGENET-A and IMAGENET-C: this could be expected since these datasets are closer to IMAGENET itself, i.e. the distribution shift is smaller, which is what nominal training optimizes for (in fact, the nominal models achieve higher accuracy than adversarially trained ones on these datasets in Fig. 6). However, in all cases there is a contribution of some of the ℓ_p-robust networks. On IMAGENET-R, IMAGENET-SKETCH and CONFLICT STIMULI, the model robust w.r.t. ℓ_∞ plays instead the most relevant role, again in line with the results in Table 1. Interestingly, in the case of CONFLICT STIMULI, the nominal classifier has a weight -0.4 (the smallest in the grid search) for all top performing soups: we hypothesize that this has the effect of reduce the texture bias typical of nominal model and emphasize the attention to shapes already important in adversarially trained classifiers. Finally, we show the composition of the soup which has the best average accuracy over all datasets (last column of Fig. 7), where the nominal and ℓ_∞-robust models have similar positive weight.

How many images does it take to find a good soup? To identify the practical limit of supervision for soup selection, we study the effect of varying the number of labelled images

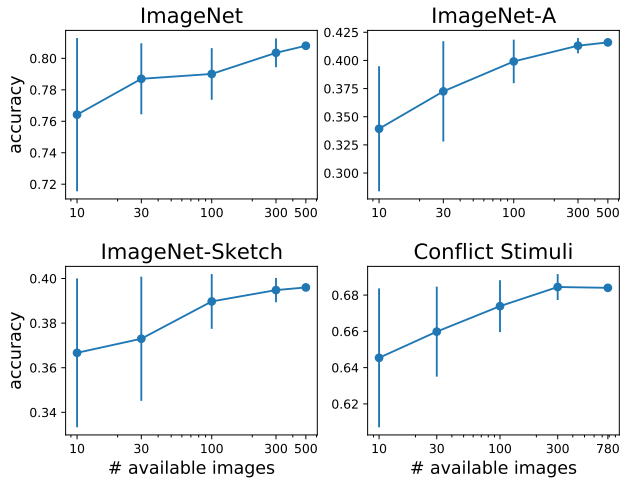

Figure 8. **Soup selection data efficiency:** we vary the number of images for selecting the best performing soups on different datasets. For each case, we plot the average accuracy on a held-out test set, and its standard deviation over 50 trials.

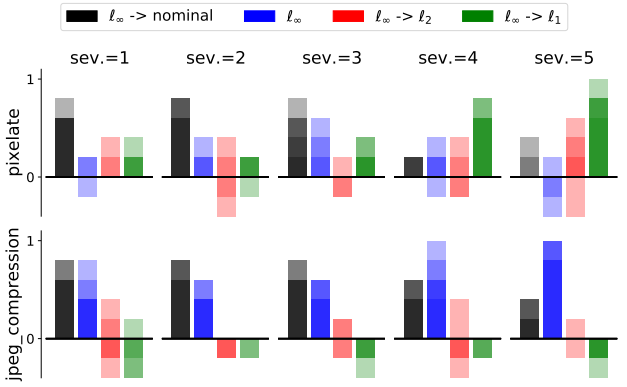

Figure 9. **Soup compositions on IMAGENET-C:** we plot the model-wise weights of the best soups across types and severities.

used to select the best soup on a new dataset. For this analysis we randomly choose 500 images from the adaptation set used for the grid search to create a held-out test set. From the remaining images, we uniformly sample k elements, select the soup which performs best on such k points, and then evaluate it on this test set. We repeat this procedure for $k \in \{10, 30, 100, 300, 500\}$ for 50 times each with different random seeds. In Fig. 8 we plot the average accuracy on the held-out test set, with standard deviation, when varying k: increasing the number of points above 100 achieves high test accuracy with limited variance. This suggests that soup selection, and thus model adaptation, can be carried out with as few as 100 examples of the new distribution.

5.2. A closer look at IMAGENET-C

While our experiments have considered IMAGENET-C as a single dataset, it consists of 15 corruptions types, each with 5 severity levels. As the various corruptions have different characteristics, one might expect the best soup to vary across them. In Fig. 9 we plot the composition of the top-5 soups for each severity level for two corruption types (as done in Fig. 7). The weights of the individual classifiers significantly change across distribution shifts: for both corruption types, increasing the severity (making perturbations stronger) leads to a reduction in the nominal weight in favor of a robust weight. However, in the case of "pixelate" the soups concentrate on the ℓ_1-robust network, while for "jpeg compression" this happens for ℓ_∞. Similar visualization for the remaining IMAGENET-C subsets are found in Fig. 11 of the Appendix. This highlights the importance of interpolating models with different types of robustness, and implies that considering each corruption type (including severity levels) as independent datasets could further improve the performance of the soups on IMAGENET-C.

6. Discussion and Limitations

Merging models with different types of robustness enables strong control of classifier performance by tuning only a few soup weights. Soups can find models which perform well even on distributions unseen during training (e.g. the IMAGENET variants). Moreover, our framework avoids co-training on multiple threats: this makes it possible to fine-tune models with additional attacks as they present themselves, and enrich the soups with them.

At the moment, our soups contain only nominal or ℓ_p-robust models, but expanding the diversity of models might aid adaptation to new datasets. We selected our soups with few-shot supervision, but other settings could potentially use soups, such as unsupervised domain adaptation [34,38], on labeled clean and unlabeled shifted data, and test-time adaptation [39,40,43], on unlabeled examples alone. Moreover, in our evaluation we have constrained the soups to belong to a fixed grid, which might miss better models: future work could develop automatic schemes to optimize the soup weights, possibly with even fewer examples, or without labeled examples (as done for test-time adaptation of non-robust models).

7. Conclusion

We show that combining the parameters of robust classifiers, without additional training, achieves a smooth trade-off of robustness in different ℓ_p-threat models. This allows us to discover models which perform well on distribution shifts with only a limited number of examples of each shift. In these ways, model soups serve as a good starting point to efficiently adapt classifiers to changes in data distributions.

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